ARJUNA JEE 2027

Lecture -12 | Mathematics

SEQUENCE & SERIES

By- Sachin Jakhar Sir



Topics *to be covered*

A **Inequality**

B **Log & Exponential Series**

C **Question Practice**

D



Question Practice

Question

[Ans. : B]

Find the sum of the series $1 \cdot 2 \cdot 3 \cdot 4 + 2 \cdot 3 \cdot 4 \cdot 5 + 3 \cdot 4 \cdot 5 \cdot 6 + \ldots\ldots$ up to n terms

A $\frac{1}{4}n(n+1)(n+2)(n+3)$

B $\frac{1}{5}n(n+1)(n+2)(n+3)(n+4)$

C $\frac{1}{5}(n-1)n(n+1)(n-2)(n+3)$

D $\frac{1}{6}n(n+1)(n+2)(n+3)(n+4)(n+5)$

$$T_n = \frac{n(n+1)(n+2)(n+3)}{5}\left((n+4)-(n-1)\right)$$

enjoy!

Question [JEE Mains-2021]

$a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \ldots = d$

Let $a_1, a_2, \ldots, a_{21}$ be an AP such that $\sum_{n=1}^{20} \frac{1}{a_n a_{n+1}} = \frac{4}{9}$. If the sum of this AP is 189, then $a_6 \cdot a_{16}$ is equal to

$$\frac{1}{a_1 a_2} + \frac{1}{a_2 a_3} + \frac{1}{a_3 a_4} + \ldots + \frac{1}{a_{20} a_{21}}$$

$$\frac{1}{d}\left(\frac{a_2 - a_1}{a_1 a_2}\right) + \frac{1}{d}\left(\frac{a_3 - a_2}{a_2 a_3}\right) + \frac{1}{d}\left(\frac{a_4 - a_3}{a_3 a_4}\right) + \ldots + \frac{1}{d}\left(\frac{a_{21} - a_{20}}{a_{21} a_{20}}\right)$$

(A) 57

(B) 72

(C) 48

(D) 36

$$\frac{1}{d}\left(\frac{\overbrace{a_2-a_1}^{d}}{a_1a_2}\right)+\frac{1}{d}\left(\frac{a_3-a_2}{a_2a_3}\right)+\frac{1}{d}\left(\frac{a_4-a_3}{a_3a_4}\right)+\cdots\cdots+\frac{1}{d}\left(\frac{a_{21}-a_{20}}{a_{21}a_{20}}\right)$$

$$T_1=\frac{1}{d}\left(\frac{1}{a_1}-\frac{1}{a_2}\right)$$

$$T_2=\frac{1}{d}\left(\frac{1}{a_2}-\frac{1}{a_3}\right)$$

$$T_3=\frac{1}{d}\left(\frac{1}{a_3}-\frac{1}{a_4}\right)$$

$$\vdots$$

$$T_{20}=\frac{1}{d}\left(\frac{1}{a_{20}}-\frac{1}{a_{21}}\right)$$

add

$$\frac{4}{9}=S_{20}=\frac{1}{d}\left(\frac{1}{a_1}-\frac{1}{a_{21}}\right)$$

$$\frac{4}{9}=\frac{1}{d}\frac{\overbrace{(a_{21}-a_1)}^{20d}}{a_{21}a_1}$$

$$\frac{4}{9}=\frac{20}{a_{21}a_1}5$$

$$a_{21}a_1=45$$

$$a_{21}=a_1+20d$$

$$a_{21}-a_1=20d$$

Question [JEE Mains-2021]

Let $a_1, a_2, \ldots, a_{21}$ be an AP such that $\sum_{n=1}^{20} \frac{1}{a_n a_{n+1}} = \frac{4}{9}$. If the sum of this AP is 189, then $a_6 \cdot a_{16}$ is equal to

84%

A 57

B 72

C 48

D 36

$a_1 + a_2 + \ldots + a_{21} = 189$

$\frac{21}{2}\left(2a_1 + (21-1)d\right) = 189$

OR

$\frac{21}{2}(a_1 + a_{21}) = 189$

$a_1 + a_{21} = 18$

$a_{21} \cdot a_1 = 45$

$a_{21} = \frac{45}{a_1}$

$a_1 + \frac{45}{a_1} = 18$

$a_1 = 3$, $a_{21} = 15$ ✓

OR $a_1 = 15$, $a_{21} = 3$ ✓

Question

Find $S = \sum_{r=1}^{\infty} \frac{r}{r^4+2r^2+9}$

$\frac{5}{24}$ #fnf.

$$\frac{r}{r^4+2r^2+9} = \frac{r}{(r^2+3+2r)(r^2+3-2r)} = \frac{1}{4}\left(\frac{(r^2+3+2r)-(r^2+3-2r)}{(r^2+3+2r)(r^2+3-2r)}\right)$$

$$T_r = \frac{1}{4}\left(\frac{1}{r^2+3-2r} - \frac{1}{r^2+3+2r}\right)$$

$$T_r = \frac{1}{4}\left(\frac{1}{r^2+3-2r} - \frac{1}{r^2+3+2r}\right)$$

$$T_1 = \frac{1}{4}\left(\frac{1}{2} - \cancel{\frac{1}{6}}\right)$$

$$T_2 = \frac{1}{4}\left(\frac{1}{3} - \cancel{\frac{1}{11}}\right)$$

$$T_3 = \frac{1}{4}\left(\cancel{\frac{1}{6}} - \cancel{\frac{1}{18}}\right)$$

$$T_4 = \frac{1}{4}\left(\cancel{\frac{1}{11}} - \cancel{\frac{1}{27}}\right)$$

$$\vdots$$

$$\infty$$

$$S_\infty = \frac{1}{4}\left(\frac{1}{2}+\frac{1}{3}\right) = \frac{5}{24}$$

$\frac{5}{24}$

#fnf.

Question [JEE Mains-2025]

If the sum of the first 10 terms of the series

$\frac{4.1}{1+4.1^4}+\frac{4.2}{1+4.2^4}+\frac{4.3}{1+4.3^4}+\cdots$ is $\frac{m}{n}$, where $\gcd(m,n)=1$, then $m+n$ is equal _____.

441

fnf

$$T_n=\frac{4n}{1+4n^4}=\frac{4n}{(1+2n^2-2n)(1+2n^2+2n)}$$

$$(1+2n^2+2n)-(1+2n^2-2n)$$

$$1^2+(2n^2)^2+2(2n^2)(1)-2(2n^2)1$$

$$(1+2n^2)^2-(2n)^2$$

$$T_n = \frac{(1+2n^2+2n)-(1+2n^2-2n)}{(1+2n^2-2n)(1+2n^2+2n)} = \frac{1}{1+2n^2-2n} - \frac{1}{1+2n^2+2n}$$

$$T_1 = \frac{1}{1} - \frac{1}{5}$$

$$T_2 = \frac{1}{5} - \frac{1}{13}$$

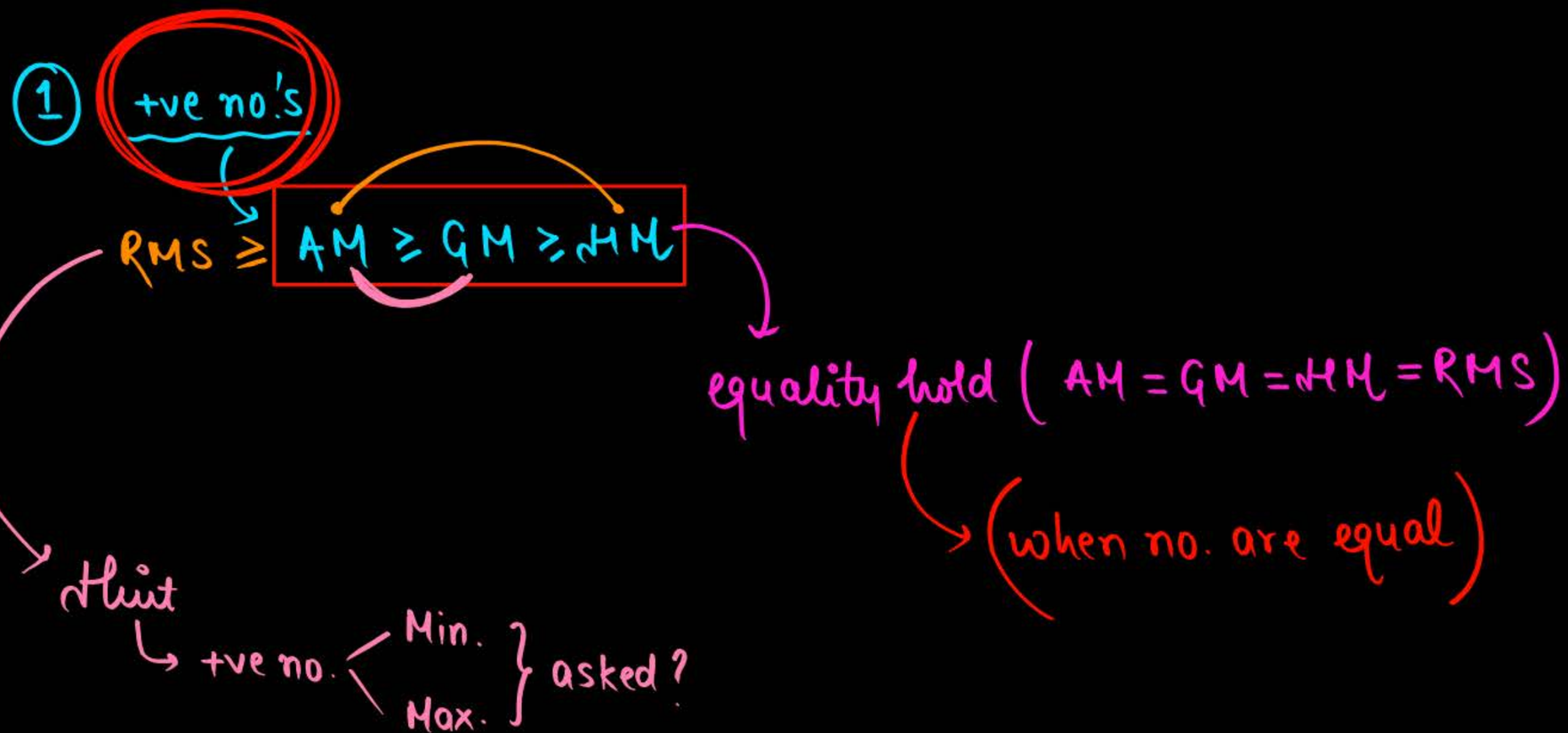
$$T_3 = \frac{1}{13} - \frac{1}{25}$$

$$\vdots$$

$$T_{10} = \frac{1}{181} - \frac{1}{221}$$

$$S_{10} = 1 - \frac{1}{221} = \frac{220}{221}$$

INEQUALITY


1
+ve no.'s
RMS ≥ AM ≥ GM ≥ HM
equality hold (AM = GM = HM = RMS)
(when no. are equal)
Hint
+ve no.
Min.
Max.
asked ?



INEQUALITIES

For n positive numbers $(a_1, a_2, \ldots. a_n)$:

R.M.S. ≥ A.M. ≥ G.M. ≥ H.M.

$$\Rightarrow \sqrt{\frac{a_1^2+a_2^2+\ldots.+a_n^2}{n}} \geq \frac{a_1+a_2+\ldots+a_n}{n} \geq (a_1 a_2 a_3 \ldots . a_n)^{1/n} \geq \frac{n}{\frac{1}{a_1}+\frac{1}{a_2}+\ldots+\frac{1}{a_n}}$$

Question

If $abcd = 1$ where a, b, c, d are positive reals then the minimum value of $a^2 + b^2 + c^2 + d^2 + ab + ac + ad + bc + bd + cd$ is

A 6

B 10

C 12

D 20

AM $\geq$ GM

$$\frac{a^2+b^2+c^2+d^2+ab+ac+ad+bc+bd+cd}{10} \geq \left(a^5 b^5 c^5 d^5\right)^{\frac{1}{9}}$$

$$\left((abcd)^5\right)^{\frac{1}{9}} = 1^{5/9} = 1$$

Acidic Question

If a, b, c are +ve numbers & $a + 2b + 3c = 7$ then find maximum value of :

(i) (ab^2c^3) (ii) $(a^2b^3c^4)$ (iii) (ab^2c)

$a, 2b, 3c \rightarrow$ +ve no.

$$\frac{\overbrace{a+2b+3c}^{7}}{3} \geq (a \times 2b \times 3c)^{\frac{1}{3}}$$

$$\frac{7}{3} \geq (6abc)^{\frac{1}{3}}$$

cube ( )^3

$$\frac{7^3}{3^3} \geq 6abc$$

$$\frac{7^3}{6 \times 3^3} \geq abc$$

$$\#\, abc|_{max.} = \frac{7^3}{6 \times 3^3}$$

Acidic Question

If a, b, c are +ve numbers & $a + 2b + 3c = 7$ then find maximum value of :

(i) $(ab^2c^3) \to \left(\frac{7}{6}\right)^6$ (ii) $(a^2b^3c^4)$ (iii) (ab^2c)

$a + 2b + 3c = 7$

$a + b + b + c + c + c = 7$

$a, b, b, c, c, c \to AM \geq GM$

$$\frac{\overbrace{a+b+b+c+c+c}^{7}}{6} \geq (a^1b^2c^3)^{\frac{1}{6}}$$

$$\frac{7}{6} \geq (a^1b^2c^3)^{\frac{1}{6}}$$

$$( )^6$$

$$\frac{7^6}{6^6} \geq a^1b^2c^3$$

$$a^1b^2c^3\Big|_{max.} = \left(\frac{7}{6}\right)^6$$

IELTS & TOEFL PREP

$a+2b+3c=7$

$a, b, b, c, c, c \rightarrow AM \geq GM$

$$\frac{7^6}{6^6} \geq a^1 b^2 c^3$$

$a=?\ \ b=?\ \ c=?$

equality hold

$a=b=b=c=c=c$

$$a+2a+3a=7$$

$$6a=7$$

$$a=\frac{7}{6}=b=c.$$

$$a^1 b^2 c^3 = \left(\frac{7}{6}\right)^1 \left(\frac{7}{6}\right)^2 \cdot \left(\frac{7}{6}\right)^3$$

$$= \left(\frac{7}{6}\right)^6$$

Acidic Question

If a, b, c are +ve numbers & $a + 2b + 3c = 7$ then find maximum value of :

(i) (ab^2c^3) (ii) $(a^2b^3c^4)$ (iii) (ab^2c)

(ii) $a + 2b + 3c = 7$

$\frac{a}{2}$ $\frac{a}{2}$ $\frac{2b}{3}$ $\frac{2b}{3}$ $\frac{2b}{3}$ $\frac{3c}{4}$ $\frac{3c}{4}$ $\frac{3c}{4}$ $\frac{3c}{4}$



(ii)

$a + 2b + 3c = 7$

$\frac{a}{2}, \frac{a}{2}, \frac{2b}{3}, \frac{2b}{3}, \frac{2b}{3}, \frac{3c}{4}, \frac{3c}{4}, \frac{3c}{4}, \frac{3c}{4}$ → AM ≥ GM

$$\frac{\frac{a}{2}+\frac{a}{2}+\frac{2b}{3}+\frac{2b}{3}+\frac{2b}{3}+\frac{3c}{4}+\frac{3c}{4}+\frac{3c}{4}+\frac{3c}{4}}{9} \geq \left(\frac{a^2 2^3 b^3 3^4 c^4}{2^2 \cdot 3^3 \cdot 4^4}\right)^{\frac{1}{9}}$$

$$\frac{\overbrace{a+2b+3c}^{7}}{9} \geq \left(\frac{3a^2b^3c^4}{2^7}\right)^{\frac{1}{9}}$$

$$\frac{7}{9} \geq \left(\frac{3a^2b^3c^4}{2^7}\right)^{\frac{1}{9}} \quad (\,)^9$$

$$\frac{7^9}{9^9} \geq \frac{3a^2b^3c^4}{2^7}$$

fnf

$$\underbrace{2^7 \cdot 7^9}_{3\times 9^9} \geq a^2b^3c^4$$

Question

If $f(x) = 2^{1+x} + 2^{1-x}$ then **minimum** value of $f(x)$ is _____ at x = _____

exponential (a^x) → always +ve

$(a = \underset{+ve}{\text{const.}},\ a \neq 1)$

$$2^{1+x} \cdot 2^{1-x} = 2^{(1+x+1-x)} = 2^2 = 4.$$

AM ≥ GM

$$\frac{2^{1+x} + 2^{1-x}}{2} \geq \left(2^{1+x} \cdot 2^{1-x}\right)^{\frac{1}{2}}$$

$$\left(2^{1+x} \cdot 2^{1-x}\right)^{\frac{1}{2}} \to 4$$

$$2^{1+x} + 2^{1-x} \geq 2 \times 4^{\frac{1}{2}}$$

$$\sqrt{4} = 2$$

$$2^{1+x} + 2^{1-x} \geq 4$$

equality: $2^{1+x} = 2^{1-x}$

$1+x = 1-x$

$2x = 0 \to x = 0$

Question

If $a, b, c > 0$ then show that : $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$

Hence deduce that for all $a_i > 0, (i = 1, 2, 3 \ldots . n)$

$$(a_1 + a_2 + a_3 + \ldots . + a_n)\left(\frac{1}{a_1}+\frac{1}{a_2}+\frac{1}{a_3}+\ldots . +\frac{1}{a_n}\right) \geq n^2$$

AM ≥ HM

$$\frac{a_1+a_2+\ldots+a_n}{n} \geq \frac{n}{\left(\frac{1}{a_1}+\frac{1}{a_2}+\ldots+\frac{1}{a_n}\right)}$$

$a, b, c \rightarrow$ AM ≥ HM

$$\frac{a+b+c}{3} \geq \frac{3}{\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)}$$

$$(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9 \checkmark$$

Question [JEE (Adv.)-2020, Paper-1]

Let m be the minimum possible value of $\log_3(3^{y_1} + 3^{y_2} + 3^{y_3})$, where y_1, y_2, y_3 are real numbers for which $y_1 + y_2 + y_3 = 9$. Let M be the maximum possible value of $(\log_3 x_1 + \log_3 x_2 + \log_3 x_3)$, where x_1, x_2, x_3 are positive real numbers for which $x_1 + x_2 + x_3 = 9$. Then the value of $\log_2(m^3) + \log_3(M^3)$ is _____.



Critical Thinking Question

New to Me → R.A.M.

If x_1, x_2 and x_3 are the positive roots of the equation $x^3 - 6x^2 + 3px - 2p = 0, p \in R$ then the value of p is _____.

$x^3 - 6x^2 + 3px - 2p = 0$ (+ − + −)

$x_1 + x_2 + x_3 = -\frac{(-6)}{1} = 6$

$x_1x_2 + x_2x_3 + x_3x_1 = +\frac{3p}{1} = 3p$

$x_1x_2x_3 = -\frac{(-2p)}{1} = 2p$

$x_1 + x_1 + x_1 = 6$

$3x_1 = 6$

$x_1 = 2$

$8 - 24 + 6p - 2p = 0$

$4p = 16$

$p = 4$.

$AM = HM = GM = 2$

$x_1 = x_2 = x_3$

$AM = \frac{x_1 + x_2 + x_3}{3} = \frac{6}{3} = 2$

$GM = (x_1x_2x_3)^{\frac{1}{3}} = (2p)^{\frac{1}{3}}$

$(2p)^{\frac{1}{3}} = 2$

$2p = 8 \rightarrow p = 4$

$HM = \frac{3}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3}} = \frac{3}{\frac{x_2x_3 + x_1x_2 + x_3x_1}{x_1x_2x_3}}$

$= \frac{3}{\frac{3p}{2p}} = \frac{3}{3} \times 2 = 2$

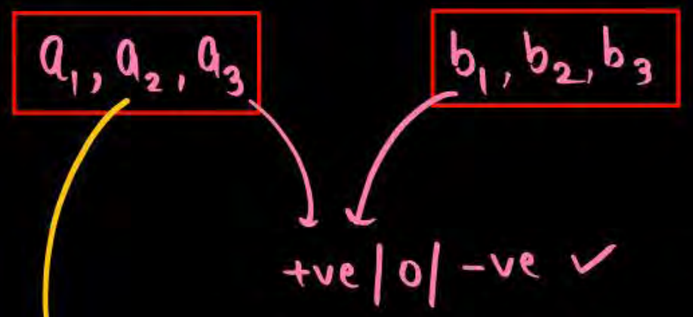


CAUCHY-SCHWARZ INEQUALITY

For real numbers: $a_1, a_2, a_3 \ldots a_n$ & $b_1, b_2, b_3 \ldots b_n$

$$(a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2 \leq (a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2)$$

Equality holds when: $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \cdots .. = \frac{a_n}{b_n}$



$$(a_1b_1 + a_2b_2 + a_3b_3)^2 \leq (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$$

$$(a_1b_1 + a_2b_2 + a_3b_3 + a_4b_4)^2 \leq (a_1^2 + a_2^2 + a_3^2 + a_4^2) \cdot (b_1^2 + b_2^2 + b_3^2 + b_4^2)$$

Proof:

$$(a_1x-b_1)^2+(a_2x-b_2)^2+(a_3x-b_3)^2 \geq 0 \quad \forall\ x\in R$$

$$a_1^2x^2+b_1^2-2a_1b_1x+a_2^2x^2+b_2^2-2a_2b_2x+a_3^2x^2+b_3^2-2a_3b_3x \geq 0 \quad \forall\ x\in R$$

$$\left(a_1^2+a_2^2+a_3^2\right)x^2-2\left(a_1b_1+a_2b_2+a_3b_3\right)x+\left(b_1^2+b_2^2+b_3^2\right)\geq 0 \quad \forall\ x\in R$$

$a_1^2+a_2^2+a_3^2$: +ve

$\Downarrow$

$D\leq 0$

$$\left(a_1b_1+a_2b_2+a_3b_3\right)^2-\left(a_1^2+a_2^2+a_3^2\right)\left(b_1^2+b_2^2+b_3^2\right)\leq 0$$

$$\left(a_1b_1+a_2b_2+a_3b_3\right)^2 \leq \left(a_1^2+a_2^2+a_3^2\right)\left(b_1^2+b_2^2+b_3^2\right)$$

equality $\Downarrow$

$a_1x-b_1=0 \rightarrow x=\frac{b_1}{a_1}$ ✓

$a_2x-b_2=0 \rightarrow x=\frac{b_2}{a_2}$ ✓

$a_3x-b_3=0 \rightarrow x=\frac{b_3}{a_3}$ ✓

a_1, a_2, a_3 $\quad$ b_1, b_2, b_3

$+ve \,|\, 0 \,|\, -ve$ ✓

$$\left(a_1b_1 + a_2b_2 + a_3b_3\right)^2 \le \left(a_1^2 + a_2^2 + a_3^2\right)\left(b_1^2 + b_2^2 + b_3^2\right)$$

$\frac{13y}{3} = 7 \Leftarrow 2\left(\frac{2y}{3}\right) + 3y = 7$

$y = \frac{21}{13}$ $\quad$ $2(x) + 3y = 7$

when: $\frac{x}{2} = \frac{y}{3}$

$x = \frac{2y}{3}$

ex: $2x + 3y = 7$ then $x^2 + y^2|_{\text{min.}} = \frac{49}{13}$

CSI $\rightarrow$ $\left(2x + 3y\right)^2 \le \left(2^2 + 3^2\right)\left(x^2 + y^2\right)$

$a_1 = 2$, $a_2 = 3$, $b_1 = x$, $b_2 = y$

$7^2 \le 13\left(x^2 + y^2\right)$

$\frac{49}{13} \le x^2 + y^2$ ✓

Question

If $3x + 4y = 1$, where $x, y \in R$, then show that $x^2 + y^2 \geq \frac{1}{25}$.

b_1 b_2

a_1 a_2

$$(a_1b_1 + a_2b_2)^2 \leq (a_1^2 + a_2^2)(b_1^2 + b_2^2)$$

$$(3x + 4y)^2 \leq (3^2 + 4^2)(x^2 + y^2)$$

1

$$\frac{1}{25} \leq x^2 + y^2$$

Question

If $x^2 + y^2 + z^2 = 5$, find maximum value of $(x + 2y + 3z)^2$.

$$(\underset{a_1}{1}\cdot\underset{b_1}{x} + \underset{a_2}{2}\cdot\underset{b_2}{y} + \underset{a_3}{3}\cdot\underset{b_3}{z})^2 \le (1^2+2^2+3^2)(x^2+y^2+z^2)$$

$$(x+2y+3z)^2 \le 14 \times 5$$

$$\le 70$$

• Ques.

$$\alpha = \left| \underset{a_1}{2} \underset{b_1}{\sin\theta} + \underset{a_2}{4} \underset{b_2}{\cos\theta} \right|_{\max} = 5.$$

$$\left(2\sin\theta + 4\cos\theta\right)^2 \le \left(3^2+4^2\right)\left(\sin^2\theta + \cos^2\theta\right)$$

$\sin^2\theta + \cos^2\theta = 1$

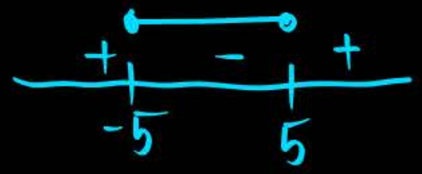
$3^2+4^2 = 25$

$$\alpha^2 \le 25$$

$$\alpha^2 - 25 \le 0$$

$$(\alpha-5)(\alpha+5) \le 0$$

$\alpha \in [-5, 5]$



$$\alpha = \left| 2\sin\theta + 4\cos\theta \right|_{\min} = -5.$$

EXPONENTIAL & LOG SERIES

Factorial:

$n! = n \times (n-1)!$

$0! = 1$

$1! = 1$

$2! = 1 \times 2 = 2$

$3! = 1 \times 2 \times 3 = 6$

$4! = 1 \times 2 \times 3 \times 4 = 24$

$5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$

$6! = 1 \times 2 \times 3 \times 4 \times 5 \times 6 = 720$

$\vdots$

$1 \times 2 \times 3 \times 4 \times 5 = 120$

$5!$ or 5

$4 = 4! = 1 \times 2 \times 3 \times 4 = 24$

$6! = (1 \times 2 \times 3 \times 4 \times 5) \times 6 = 720$

$6! = 5! \times 6$

- $n! = n \times (n-1)!$
- $6! = 6 \times 5!$
- $18! = 18 \times 17!$

$\frac{6!}{4!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1} = 30.$

$\frac{11!}{8!} = \frac{11 \times 10 \times 9 \times 8!}{8!} = 990.$

DO IT BY YOURSELF (DIBY)

TELESCOPIC METHOD OF DIFFERENCE

DIBY-88

$\frac{1}{3^2-1}+\frac{1}{5^2-1}+\frac{1}{7^2-1}+\cdots+\frac{1}{(201)^2-1}$ is equal to [JEE Mains-2021]

(A) $\frac{101}{404}$ (B) $\frac{25}{101}$ (C) $\frac{101}{408}$ (D) $\frac{99}{400}$

[Ans. : (B)]

DIBY-89

If the sum of series $\frac{1}{1\cdot(1+d)}+\frac{1}{(1+d)(1+2d)}+\cdots+\frac{1}{(1+9d)(1+10d)}$ is equal to 5, then $50d$ is equal to: [JEE Mains-2024]

(A) 20 (B) 5 (C) 15 (D) 10

[Ans. : (A)]

DIBY-90

The sum $\sum_{n=1}^{21}\frac{3}{(4n-1)(4n+3)}$ is equal to [JEE Mains-2022]

(A) $\frac{7}{87}$ (B) $\frac{7}{29}$ (C) $\frac{14}{87}$ (D) $\frac{21}{29}$

[Ans. : (B)]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-91

The sum of 10 terms of the series $\frac{3}{1^2\times2^2}+\frac{5}{2^2\times3^2}+\frac{7}{3^2\times4^2}+\cdots$ is [JEE Mains-2021]

(A) $\frac{143}{144}$ (B) 1 (C) $\frac{99}{100}$ (D) $\frac{120}{121}$

[Ans. : (D)]

DIBY-92

If $\frac{1}{2\times3\times4}+\frac{1}{3\times4\times5}+\frac{1}{4\times5\times6}+\cdots+\frac{1}{100\times101\times102}=\frac{k}{101}$, then $34k$ is equal to

[JEE Mains-2022]

[Ans. : 286]

DIBY-93

If $\sum_{k=1}^{10}\frac{k}{k^4+k^2+1}=\frac{m}{n}$, where m and n are co-prime, then $m+n$ is equal to

[JEE Mains-2022]

[Ans. : 166]

TELESCOPIC METHOD OF DIFFERENCE

DIBY-94

The sum to 10 terms of the series $\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+\cdots$ is

[JEE Mains-2023]

(A) $\frac{59}{111}$ (B) $\frac{55}{111}$ (C) $\frac{56}{111}$ (D) $\frac{58}{111}$

[Ans. : (B)]

DIBY-95

If $a_n=\frac{-2}{4n^2-16n+15}$, then $a_1+a_2+\cdots+a_{25}$ is equal to: [JEE Mains-2023]

(A) $\frac{51}{144}$ (B) $\frac{49}{138}$ (C) $\frac{50}{141}$ (D) $\frac{52}{147}$

[Ans. : (C)]

DIBY-96

The sum of the series $\frac{1}{1-3\cdot1^2+1^4}+\frac{2}{1-3\cdot2^2+2^4}+\frac{3}{1-3\cdot3^2+3^4}+\ldots$ up to 10 terms is

[JEE Mains-2024]

(A) $\frac{45}{109}$ (B) $-\frac{45}{109}$ (C) $\frac{55}{109}$ (D) $-\frac{55}{109}$

[Ans. : (D)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Only 2 Exercise

Topicwise & Bridge(Setu)- 4 sittings each of 1 HR

Next # Chapter
↳ Trigo !!

It is never about end result

It's always about Journey

Thank You

#futureIITians